

MULTI-PARAMETRIC ASSESSMENT OF DRINKING WATER SAFETY AND HOUSEHOLD PRACTICES IN ISALE-OYO COMMUNITY, SOUTHWESTERN NIGERIA.

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ABSTRACT

Access to safe drinking water remains a critical public health challenge in indigenous urban settlements across sub-Saharan Africa, where sanitation deficits, aging infrastructure, and poor household water management compound source-level contamination. Isale-Oyo indigenous community in Oyo Town, southwestern Nigeria, exemplifies these vulnerabilities through high population density, traditional compound housing, shared sanitation facilities, and near-total dependence on shallow groundwater. This study assessed household water-handling, storage, treatment practices; physicochemical and bacteriological water quality; and some heavy metal concentrations in drinking water sources within Isale-Oyo, benchmarked against WHO (2022) and Nigerian Standard for Drinking Water Quality (NSDWQ, 2015) limits. A cross-sectional descriptive design was used. Structured questionnaires were administered to 90 households, and 30 water sources (hand-dug wells and boreholes) were sampled in duplicate across GPS-verified sites. Physicochemical parameters followed APHA (2022) methods; microbiological analysis quantified total coliform count, *Escherichia coli*, and total heterotrophic bacterial count; iron and lead were quantified. Data were analysed using descriptive statistics, Pearson correlation, one-way ANOVA, and chi-square ($p < 0.05$). Mean pH (5.66 ± 0.50) fell below WHO range, while chloride, phosphate, and iron exceeded permissible limits. All samples contained detectable *E. coli* and coliforms, with strong inter-indicator correlations confirming pervasive faecal contamination ($r = 0.840$, $p < 0.001$). Lead remained within limits. Notably, 45.6% of respondents never treated their water, 50% practiced open defecation, and 67.8% used open dumping. Findings indicate bacteriologically unsafe, chemically compromised water driven by poor sanitation and household practices, warranting urgent well protection, treatment promotion, and sanitation interventions.

.Keywords: *Drinking water quality, Groundwater contamination, Heavy metals; Household water practices, Microbiological indicators.*

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INTRODUCTION

Water permeates all things humans; health, dignity, economic opportunity but access to clean water is one of the biggest challenges in the 21st century. According to the World Health Organization (WHO) and the United Nations Children's Fund (UNICEF) estimates in 2022, there were 2.2 billion people in the world without safely managed drinking water with 703 million having unimproved drinking water (WHO and UNICEF, 2023). The impact of this inequity on human lives is starkly illustrated by mortality: approximately 1.4 million people die each year due to water-related disease, which is overwhelmingly diarrhoeal and disproportionately affects children under five years of age (WHO, 2023). Even by 2024, the global estimates indicate that just 2.1 billion more people, on top of the already achieved, had access to safely managed services (UNICEF, 2025), a sobering reminder that progress is still far from universal.

The safety or danger of water is seldom a direct result of a single driver of change – rather it is the combination of forces, both natural and human. The geology and hydrogeological gradients, climate variability, and seasonal rainfall pattern influence water quality results in the context of urbanization, industrial discharge, agricultural runoff, and poor sanitation practices. Groundwater, accounting for almost half the world's potable requirements (WWQA 2024), has long been believed to be the safer alternative, shielded in theory by the natural filtration in soil layers. However, this protection is being depleted. Pathogenic microorganisms, nitrates, heavy metals, and anthropogenic pollutants are invading aquifers which were once considered secure in high-density areas (Taiwo *et al.*, 2023). The situation in Sub-Saharan Africa shows this erosion starkly: in 2024, more than 400 million people in Sub-Saharan Africa were without safely managed water (UNICEF, 2024); and waterborne disease like cholera, typhoid or dysentery continues to overwhelm already-challenged health systems (Dinka and Nyika, 2024).

Nigeria is an embodiment of this continental crisis in miniature. On the surface, statistics tell us that about 70% of households have access to improved drinking water sources (NBS, 2023) which seems a comforting number at first glance. It masks a clear urban-rural disparity, however, as peri-urban and indigenous communities continue to rely on hand-dug wells, boreholes, springs and vendors. The situation deteriorated to a tragic level in 2023, when thousands of suspected cases of cholera were reported in various states of Nigeria (NCDC, 2024), highlighting the extent of water safety failures and the inadequacy of existing interventions.

Oyo State is an example of groundwater dependent urbanization, whereby the failure of urban water supply systems has led communities to relying on private boreholes and hand dug wells as their natural alternatives. Both geology and human activities contribute to the variability in water quality; it is also known that the closer a study site is to a waste disposal site, the higher the likelihood of microbial contamination (Ganiyu *et al.*, 2021; Balogun *et al.*, 2022). Historically, Oyo Town, an oldest urban settlement of Yorubaland, is the capital of the ancient Yoruba kingdom of Oyo, whose population has increased beyond the capacity of its environmental infrastructure. The inadequate sewerage projects, poor drainage and refuse disposal practices, weak sanitation systems, and the proximity of sanitation facilities and water body to drinking water in the town have all combined to pose a risk to drinking water safety (Balogun *et al.*, 2022). This vulnerability is most pronounced in Isale-Oyo, an indigenous community with an exceptionally high population density, common sanitation facilities, traditional compound housing, near-obsolescence of utilities, and dependence on shallow groundwater. Wells and boreholes are located dangerously near septic tanks, soak-away pits, open drains, and refuse heaps, allowing contamination to seep through the soil

by runoff, leaching and lateral seepage, particularly during the rainy season (Taiwo *et al.*, 2023). The risks of chronic low-level exposure are also substantial: constituent metals like lead and iron are not biodegradable and bioaccumulate, meaning that they can cause neurological, renal and systemic damage over time (WHO, 2022).

Contaminating at the source is only part of the problem. Water may also become unsafe before it is even used, either during transport to the village, in the storage tank where it is not stored in hygienic containers, or at the household level where it is not treated (Odukoya and Ifarajinmi, 2021; Nfor *et al.*, 2025). The Household Water Treatment and Safe Storage (HWTS) framework developed by the World Health Organization (WHO) is an effective countermeasure that can reduce the incidence of diarrhoeal disease by up to 45% when implemented properly; however, its adoption is inconsistent throughout Nigeria's low-income communities (Atalay *et al.*, 2024). However, to date, no study has considered Isale-Oyo from an integrated perspective by analyzing the physicochemical quality, microbiological safety, heavy metal contamination and household water handling practices. Studies conducted in the past in Oyo State have focused either on the physicochemical or the microbiological aspect, but not on the behavioral factors that influence the real risk associated with the use of the point of use. This fragmentation has real implications: if a broad contaminant profile isn't associated with the human causes of the contamination, there is no evidence base for public health officials to use in designing effective interventions that are targeted, not generic. The present study explores this gap, providing a comprehensive evaluation of physicochemical parameters, microbiological indicators, selected concentrations of heavy metals and household water-handling practices all compared with the permissible limits established by WHO (2022).

METHODOLOGY

Research Design

A cross-sectional descriptive design complemented by laboratory-based analytical procedures was adopted. The cross-sectional design allowed for simultaneous collection of data on drinking water quality and household practices at a specific time point and was able to provide a representative picture of existing conditions at that time point (Setia, 2016). This design has been extensively used in similar research (Creswell and Creswell, 2018; Taiwo *et al.*, 2023; Terefe *et al.*, 2024). The laboratory part involved quantitative analytical means to characterize the physicochemical, microbiological and heavy metal parameters of water samples collected. A correlational analytical component examined the relationship between the water quality parameters and household practices using inferential statistics.

Study Area

The study was carried out in the indigenous community of Isale-Oyo in the Oyo Town of Atiba Local Government Area in Oyo State, southwest Nigeria (Figure 1). Oyo Town lies approximately between latitude 7.85°N–7.95°N and longitude 3.88°E–4.02°E and is one of the major urban centres in the state (NPC, 2023). Oyo Town is an ancient city with high population density and traditional settlement patterns that is one of the most historically important Yoruba cities in Nigeria. Isale-Oyo is a densely populated indigenous neighborhood characterized by traditional compound housing with multiple families sharing residential courtyards, narrow access roads, aging utility infrastructure, and predominantly shared or communal sanitation facilities. The climate of the community is tropical humid with two seasons: rainy during April to October (highest during June and July) and dry during November to March, having a mean annual rainfall of about 1,200 –

1,500 mm (NiMet, 2024). Generally, favorable hydrogeologic conditions allow shallow unconfined aquifers to be exploited by hand-dug wells, which are usually 5–20 m deep.

The water supply from the public water supply system is not regular and sufficient in the community and most people are using water from hand dug wells and boreholes for domestic water use. Sanitation infrastructure comprises predominantly traditional pit latrines, septic tanks and soak-away pits, with a large number of people practicing open defecation. The main methods of waste management are municipal collection (irregular), open dumping and open burning in and around residential areas. Groundwater under these environmental conditions is more susceptible to physicochemical, microbiological and heavy metal contamination (Ganiyu *et al.*, 2021; Taiwo *et al.*, 2023).

Study Population and Sampling

The study population was made up of the households in the indigenous community of Isale-Oyo with the primary emphasis on the primary caregivers who collect, store and manage water of the households. Households with regular access to a primary water source were included. A multistage sampling technique was used. The community was purposefully chosen due to its groundwater dependence and environmental vulnerability profile. The sample size of 90 households was selected using systematic random sampling technique from household registries of the community for administration of questionnaires, which is similar to the sample size in other water quality studies conducted in the community of Nigeria (Ganiyu *et al.*, 2021; Nfor *et al.*, 2025). To assess water quality, 30 drinking water sources (hand dug wells and boreholes) were stratified randomly sampled throughout the community ensuring spatial representation. Each of the 30 sources was sampled in duplicate, yielding 60 water samples in total. The location (GPS) was taken for all 30 sampling points to enable spatial mapping and documentation (Table 1).

Table 1: GPS Coordinates of Drinking Water Sampling Locations in Isale-Oyo Indigenous Community of Oyo Town.

Sample	Latitude (°N)	Longitude (°E)	Sample	Latitude (°N)	Longitude (°E)
S1	7.8642	3.9275	S16	7.8623	3.9299
S2	7.8813	3.9334	S17	7.8821	3.9325
S3	7.8831	3.9329	S18	7.8799	3.9233
S4	7.8514	3.9497	S19	7.8637	3.9267
S5	7.8921	3.8998	S20	7.8633	3.9258
S6	7.9127	3.8725	S21	7.7995	3.9183
S7	7.9463	3.9453	S22	7.8915	3.9312
S8	7.8990	3.9563	S23	7.8864	3.9305
S9	7.8865	3.8989	S24	7.8854	3.9317
S10	7.8665	3.9268	S25	7.8647	3.9269
S11	7.8497	3.9324	S26	7.8596	3.9185
S12	7.8692	3.8978	S27	7.8588	3.9174
S13	7.8911	3.9327	S28	7.8832	3.9296
S14	7.8791	3.9263	S29	7.8826	3.9274
S15	7.8769	3.9319	S30	7.8813	3.9312

Source: GPS field survey, 2026.

Data Collection

Questionnaire Administration

Face-to-face interviews were conducted with 90 household heads or designated adult water managers using a structured, pre-test questionnaire. The instrument measured socio-demographic information (sex, age, marital status, education, occupation and income), main source of water, water handling and storage, water treatment, frequency of treatment, behavioral aspects of hygiene and sanitation (use of toilets, hand washing, waste disposal), and history of water-related illness self-reported.

Water Sample Collection

Water Samples were taken according to the standard APHA (2022) method. Samples for microbiological analysis were taken in pre-sterilized 250 mL borosilicate glass bottles. Polyethylene bottles (500mL) were employed for physicochemical and heavy metal analysis and were acid-washed (10% HNO₃) and triple rinsed with sample water before use. Samples for microbiology were collected in an aseptically manner, and flushed for 2 minutes using well pumps and borehole taps. All samples were shipped in coolers with ice for microbiological and physicochemical parameters, which were analyzed within 6 hours and 24 hours, respectively, of being collected, according to the holding times outlined by APHA (2022).

Laboratory Analytical Procedures

The pH and temperature were measured in the field with calibrated portable pH/temperature meter (HANNA Instruments HI9813-6). A calibrated turbidimeter was used for measuring turbidity. Electrical Conductivity (EC) was measured by using a conductivity meter and TDS was calculated as $EC \times 0.64$ (Rusydi, 2018). The standard titration and colorimetric methods were used to obtain data on alkalinity (total), chloride, hardness, sulphate, phosphate, nitrate, and acidity as described by the American Public Health Association (APHA) (2022). All measurements were taken twice and the means were calculated.

Total coliform count was carried out by membrane filtration using m-Endo agar and incubated at 35°C for 24 hours. The *Escherichia coli* (EC) and faecal coliform (FC) counts were determined by growing EC agar at 44.5°C for 24 hours. While FC comprises all thermotolerant coliforms capable of growth at 44.5°C, *E. coli* represents a specific, more definitive subgroup within this population and was used in the present study as the more precise indicator of recent faecal contamination. The total heterotrophic bacterial count (THBC) was carried out by pour plate method using nutrient agar and incubated at 37°C for 24–48 hrs (APHA, 2022). Suspended solids were determined gravimetrically by filtering a known volume of each water sample through a pre-weighed 0.45 µm membrane filter, drying the residue to constant weight at 103–105°C, and calculating the mass difference per litre of sample (APHA, 2022). Each microbiological analysis had been performed three times and the results presented as colony forming units per millilitre (cfu/mL) $\times 10^4$.

Samples for heavy metal content were preserved in 2 mL of concentrated HNO₃ /100 mL of sample. Acid digestion was achieved by heating gently with a mixture of concentrated HNO₃ and H₂SO₄ (3:1 v/v). Iron (Fe) and lead (Pb) concentrations were determined by flame Atomic Absorption Spectrophotometry (AAS, PerkinElmer Analyst 400) using standard procedures recommended by APHA (2022). Certified standard solutions (1,000 mg/L, Merck) were used to prepare a calibration curve. The method detection limits were 0.005 mg/L for Fe and 0.001 mg/L for Pb. Quality assurance procedures employed were the use of certified reference materials, duplicate analyses and reagent blanks.

Compliance and Ethical Considerations

Prior to data collection, institutional ethical approval was obtained from Ethical Review committee of Ajayi Crowther University (FNS/ERC/2025/0791DWS). All participants gave informed consent prior to the administration of the questionnaire. Participants' information was kept anonymous and confidential during the study. The participation was voluntary, and there was no requirement to continue participation. Water samples were taken according to biosafety measures to ensure the safety of the field researchers.

Data Analysis

Data were organized on Ms. Excel 2021 and analysed with IBM SPSS Statistics version 27. Socio-demographic variables, household practices and water quality parameters were presented as descriptive statistics (frequencies, percentages, mean, SD). The results of physicochemical, microbiological and heavy metal were systematically compared with the permissible limits of WHO (2022). Pearson product-moment correlation was used to explore correlations between physicochemical and microbiological parameters. The associations between household practices and water quality outcomes were measured using chi-square tests. One-way analysis of variance (ANOVA) was used to compare the mean parameter values among the water source types. The tests were all done at 95% confidence level ($\alpha = 0.05$).

RESULTS

Socio-Demographic Characteristics of Respondents

A total of 90 questionnaires were completed (100% response rate). The majority (57.8%) of the respondents were females while the males made up the remaining 42.2%, reflecting the fact that females as the main water managers in the communities are predominant in these communities. The largest age cohort was 26–35 years ($n = 34$; 37.8%), followed by 36–45 years ($n = 22$; 24.4%), 18–25 years ($n = 20$; 22.2%), and ≥ 46 years ($n = 14$; 15.6%). The majority of respondents were married ($n = 50$; 55.6%) with single (33.3%) and others (widowed, separated) respondents at 11.1%. The predominant educational level was secondary ($n = 36$; 40.0%), followed by tertiary (24.4%), primary (22.2%), and no formal education (13.3%). The major occupations were traders (35.6%), artisans (22.2%) and civil servants (20.0%). Monthly income was broadly distributed: ₦50,000–₦100,000 (37.8%), <₦50,000 (31.1%), and >₦100,000 (31.1%). A stable long-term resident population was indicated by the majority (46.7%) having lived in the community for over 10 years (Table 2).

Household Water Sources, Handling, Storage, and Treatment Practices

Groundwater was the main source of drinking water for all respondents (100%), with 72.2% ($n = 65$) using water from a hand dug well, and 27.8% ($n = 25$) using water from a borehole. The most common treatment method reported by the respondents who reported treating their water was boiling (54.4%), followed by filtration (22.2%), chlorination (17.8%) and sedimentation (5.6%). Most importantly, 45.6% said that they never treated their drinking water before they drank it and 34.4% treated water consistently (Table 3). It is especially concerning that almost half of the respondents do not treat their drinking water at all, in light of the microbiological contamination found. Concerning storage, 84.4% used covered containers, which is a relatively positive storage practice, while 15.6% stored water in uncovered containers, which put stored water at risk of environmental contamination.

Table 2: Socio-Demographic Characteristics of Respondents (n = 90)

Variable	Category	Frequency	Percentage (%)
Gender	Male	38	42.2
	Female	52	57.8
Age	18–25 years	20	22.2
	26–35 years	34	37.8
	36–45 years	22	24.4
	≥46 years	14	15.6
Marital Status	Single	30	33.3
	Married	50	55.6
	Others	10	11.1
Education	No formal education	12	13.3
	Primary	20	22.2
	Secondary	36	40.0
	Tertiary	22	24.4
Occupation	Civil servant	18	20.0
	Trader	32	35.6
	Artisan	20	22.2
	Others	20	22.2
Monthly Income	< ₦50,000	28	31.1
	₦50,000–₦100,000	34	37.8
	> ₦100,000	28	31.1
Years in Community	1–5 years	18	20.0
	6–10 years	30	33.3
	> 10 years	42	46.7

Source: Field Survey, 2026.

Table 3: Household Water Sources and Water-Handling Practices (n = 90)

Variable	Category	Frequency	Percentage (%)
Water Source	Hand-dug well	65	72.2
	Borehole	25	27.8
Treatment Method	Filtration	20	22.2
	Boiling	49	54.4
	Chlorination	16	17.8
	Sedimentation	5	5.6
	Always	31	34.4
Treatment Frequency	Sometimes	18	20.0
	Never	41	45.6
Storage Practice	Covered container	76	84.4
	Uncovered container	14	15.6

Source: Field Survey, 2026.

Hygiene and Sanitation Practices

Sanitation results were a serious concern. Open defecation was the predominant sanitation practice (50.0%; n = 45), followed by pit latrine use (32.2%; n = 29), water closet systems (13.3%; n = 12), and ventilated pit latrines (4.4%; n = 4). Densely populated indigenous population provides direct and pervasive routes of fecal contamination of shallow groundwater through open defecation. Hand washing behaviour was also sub-optimal with 74.4% washing their hands with water alone and only 25.6% washing water and soap (a critical gap in infection prevention). Table 4 shows that 67.8% (n = 61) of the respondents disposed of their waste by open dumping, while 32.2% (n = 29) did so by burning, and none of them used municipal waste collection services.

Table 4: Hygiene and Sanitation Practices among Respondents (n = 90)

Variable	Category	Frequency	Percentage (%)
Sanitation Facility	Water closet	12	13.3
	Ventilated pit latrine	4	4.4
	Pit latrine	29	32.2
	Open defecation	45	50.0
Hand Washing	Water only	67	74.4
	Water and soap	23	25.6
Waste Disposal	Open dumping	61	67.8
	Burning	29	32.2

Source: Field Survey, 2026.

Self-Reported Water-Related Illness

Table 5 shows the health symptoms reported by respondents. Typhoid fever and itchy skin were the most frequently reported conditions (56.7% each; n = 51), followed by cholera (47.8%; n = 43), diarrhoea (41.1%; n = 37), mottled skin (36.7%; n = 33), skin infections (32.2%; n = 29), and fever (26.7%; n = 24). The high prevalence of symptoms attributable to typhoid and cholera strongly suggests that there is continued fecal–oral transmission via contaminated water and food. Notably, 37.8% of respondents reported frequent occurrence of water related health problem and 34.4% reported occasional occurrence, with only 27.8% reporting rare occurrence meaning there is no episodic nature of exposure to water borne health hazards but rather there is an endemic nature within the community.

Table 5: Self-Reported Water-Related Health Symptoms and Illness Frequency (n = 90)

Variable	Category	Frequency	Percentage (%)
Water-Related Illness	Fever	24	26.7
	Cholera	43	47.8
	Typhoid	51	56.7
	Diarrhoea	37	41.1
	Skin infection	29	32.2
	Mottled skin	33	36.7

Variable	Category	Frequency	Percentage (%)
Illness Frequency	Itchy skin	51	56.7
	Frequently	34	37.8
	Occasionally	31	34.4
	Rarely	25	27.8

Source: Field Survey, 2026.

Note: Multiple responses permitted for illness type.

Physicochemical Quality of Drinking Water

Descriptive statistics of all the physicochemical parameters determined in the 30 drinking water sources sampled ($n = 30$ samples, analysed in duplicate) are presented in Table 6 and Figure 1. All the drinking water sources sampled were acidic with a mean pH of 5.66 ± 0.50 (range of 4.91–6.22), which is below the permissible range of pH (6.5–8.5) set by WHO. Temperature (mean $27.23 \pm 0.33^\circ\text{C}$; range 26.50 – 28.05°C) was not of safety concern as it was within the ambient tropical range. Electrical conductivity (EC mean value $151.59 \pm 4.99\mu\text{S/cm}$) and TDS (mean value $0.53 \pm 0.09\text{ mg/L}$) values were found to be in the range of WHO permissible limits for EC ($1,000\mu\text{S/cm}$) and TDS (500 mg/L), respectively, indicating moderate loading of ions without gross mineralization. All the samples had turbidity values below the WHO guideline of 5 NTU (mean $0.91 \pm 0.18\text{ NTU}$) but should not be regarded as microbiologically safe as the turbid water does not imply contaminated water and vice versa.

The alkalinity was within permissible limit of WHO (500 mg/L) with mean value of ($278.02 \pm 92.99\text{ mg/L}$) Range (108.55 – 505.40 mg/L) which showed that the buffering capacity is satisfactory. Sulphate (mean $10.55 \pm 2.10\text{ mg/L}$) and nitrate (mean $0.12 \pm 0.02\text{ mg/L}$) were far lower than the standards set by WHO (250 mg/L and 50 mg/L respectively), indicating that there was little contamination with sulphate and low levels of nitrate in the samples sources, which is somewhat surprising given the high levels of open defecation and open dumping observed in the community, and the efficient leaching process, which dilutes nitrate in the relatively humid groundwater environment.

However, the chloride concentration (mean $432.96 \pm 118.65\text{ mg/L}$; range 106.65 – 639.05 mg/L) was significantly higher than the aesthetic guideline set by WHO of 250 mg/L , where the peak of 639.05 mg/L was over 2.5 times the guideline. Phosphate ($8.08 \pm 1.18\text{ mg/L}$, range 6.45 – 10.11 mg/L) was found to be higher than the recommended value by WHO (5.0 mg/L) in all samples. There was no WHO guideline value for acidity but it was similar to the observed pH values of the water samples, which ranged from 0.018 to 0.900 mg/L, with a mean of $0.103 \pm 0.152\text{ mg/L}$.

Table 6: Descriptive Statistics of Physicochemical Parameters of Drinking Water Samples from Isale-Oyo Indigenous Community ($n = 30$)

Parameter	Min	Max	Mean $\pm$ SD	WHO Limit	Status
pH	4.91	6.22	5.66 ± 0.50	6.5–8.5	Non-compliant
Temperature ($^\circ\text{C}$)	26.50	28.05	27.23 ± 0.33	Ambient	Compliant
EC ($\mu\text{S/cm}$)	145.55	160.45	151.59 ± 4.99	1000	Compliant
Turbidity (NTU)	0.65	1.20	0.91 ± 0.18	5.0	Compliant
TDS (mg/L)	0.31	0.70	0.53 ± 0.09	500	Compliant

Parameter	Min	Max	Mean $\pm$ SD	WHO Limit	Status
Alkalinity (mg/L)	108.55	505.40	278.02 $\pm$ 92.99	500	Compliant
Chloride (mg/L)	106.65	639.05	432.96 $\pm$ 118.65	250	Non-compliant
Acidity (mg/L)	0.018	0.900	0.103 $\pm$ 0.152	—	—
Sulphate (mg/L)	7.45	16.59	10.55 $\pm$ 2.10	250	Compliant
Phosphate (mg/L)	6.45	10.11	8.08 $\pm$ 1.18	5.0	Non-compliant
Nitrate (mg/L)	0.09	0.17	0.12 $\pm$ 0.02	50	Compliant

Source: Laboratory Analysis, 2026. WHO limits from WHO (2022) Guidelines for Drinking-water Quality.

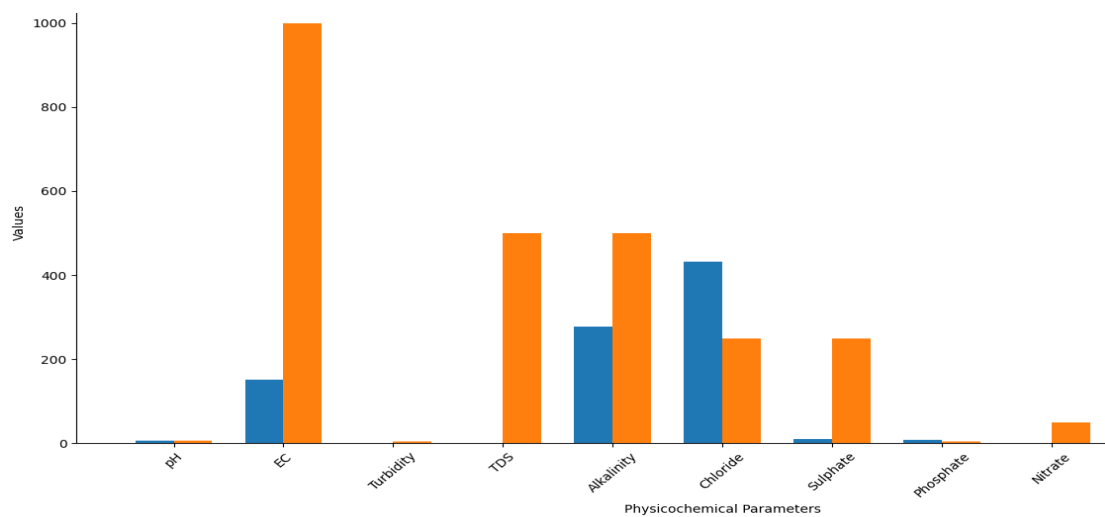


Figure 1: Comparison of Physicochemical Parameters of Drinking Water Samples with WHO Permissible Standards
Microbiological Quality of Drinking Water

The microbiological analysis results are given in Table 7. All 30 sampled sources yielded detectable total coliforms (TCC: mean $0.53 \pm 0.13 \times 10^4$ cfu/mL; range 0.25–0.80), *E. coli* (mean $0.31 \pm 0.07 \times 10^4$ cfu/mL; range 0.20–0.45), and fecal coliforms (FCC: mean $0.15 \pm 0.05 \times 10^4$ cfu/mL; range 0.10–0.20). Total heterotrophic bacterial count (THBC) was universal ($0.62 \pm 0.07 \times 10^4$ cfu/mL, range 0.50–0.75), which showed that total heterotrophic bacteria grew in the groundwater environment. Suspended solid counts were measurable in about half of the samples with a mean of $0.09 \pm 0.08 \times 10^4$ cfu/mL (range 0.00–0.20). This is a categorical compliance failure on the WHO guidelines as *E. coli* has been detected in all the samples with a guideline of zero *E. coli* in 100 mL of drinking water (WHO, 2022).

Table 7: Descriptive Statistics of Microbiological Parameters of Drinking Water Samples (n = 30)

Parameter	Min ($\times 10^4$ cfu/mL)	Max ($\times 10^4$ cfu/mL)	Mean $\pm$ SD ($\times 10^4$ cfu/mL)	WHO Limit	Status
Total Coliform Count	0.25	0.80	0.53 $\pm$ 0.13	0 /100 mL	Non-compliant
<i>E. coli</i> Count	0.20	0.45	0.31 $\pm$ 0.07	0 /100 mL	Non-compliant

Parameter	Min ($\times 10^4$ cfu/mL)	Max ($\times 10^4$ cfu/mL)	Mean $\pm$ SD ($\times 10^4$ cfu/mL)	WHO Limit	Status
Faecal Coliform Count	0.10	0.20	0.15 ± 0.05	0 /100 mL	Non-compliant
THBC	0.50	0.75	0.62 ± 0.07	—	—
Suspended Solids	0.00	0.20	0.09 ± 0.08	—	—

Source: Laboratory Analysis, 2026. THBC = Total Heterotrophic Bacterial Count.

Heavy Metal Concentrations

Concentrations of total hardness, iron, and lead are shown in table 8. Total hardness was generally low (mean 7.52 ± 1.50 mg/L; range 3.44–10.50 mg/L) and within the permissible limits of 500 mg/L set by WHO which further indicate soft water conditions, as found in the study area, due to the acidic pH and Precambrian basement geology. The sampling range for iron concentration was 0.31–0.72 mg/L, with a mean level of 0.50 ± 0.09 mg/L; this was above the WHO guideline of 0.30 mg/L and the highest concentration recorded (0.72 mg/L) was more than double the WHO guideline. 30 of all sources failed to meet the WHO iron standard. The mean concentration of lead in the drinking water samples (0.0015 ± 0.0007 mg/L) was not at an acute concern level; the range of lead concentrations in drinking water from 0.001 to 0.003 mg/L was still below the WHO guideline of 0.01 mg/L. Any non-detectable level of lead in drinking water is of epidemiological concern, as there is no safe threshold for neurological effects in children (WHO, 2022).

Table 8: Heavy Metal and Total Hardness Parameters of Drinking Water Samples (n = 30)

Parameter	Min (mg/L)	Max (mg/L)	Mean $\pm$ SD (mg/L)	WHO Limit (mg/L)	Status
Total Hardness	3.44	10.50	7.52 ± 1.50	500	Compliant
Iron (Fe)	0.31	0.72	0.50 ± 0.09	0.30	Non-compliant
Lead (Pb)	0.001	0.003	0.0015 ± 0.0007	0.01	Compliant

Source: Laboratory Analysis, 2026. WHO limits from WHO (2022).

Pearson Correlation Analysis of Physicochemical and Microbiological Parameters

Table 9 shows the Pearson correlation matrix of selected parameters. The total hardness was positively correlated with FCC ($r = 0.362$; $p = 0.049$), which indicated that there might be a common source of contamination for hardness-inducing minerals and fecal organisms, likely from the surrounding soil matrix. Iron showed a significant negative correlation with FCC ($r = -0.461$; $p = 0.010$), which may be related to the geochemical competitive dynamics occurring between the iron deposition zones and the fecal contamination pathway in the aquifer. There were no significant correlations between lead and any microbiological indicator, in line with its geogenic source, as opposed to fecal contamination ($p > 0.05$ throughout).

The correlation between TCC and *E. coli* was strong and positive ($r = 0.840$; $p < 0.001$) as *E. coli* represents a significant component of total coliforms, and between TCC and FCC was also strong and positive ($r = 0.602$; $p < 0.001$) reflecting similar fecal sources. The significant correlation between TCC and THBC ($r = 0.374$; $p = 0.041$) and between TCC and SS Count ($r = 0.470$; $p = 0.009$) indicate that particulate matter in water can be a source of attachment surface and nutrients to

support the growth of coliform. *E. coli* was significantly correlated with both FCC ($r = 0.734$; $p < 0.001$) and THBC ($r = 0.499$; $p = 0.005$); and FCC with THBC ($r = 0.448$; $p = 0.013$), indicating a similar signal for fecal contamination of these groundwater samples for all microbiological indicators.

Table 9: Pearson Correlation Analysis of Selected Physicochemical and Microbiological Parameters

Variable Pair	r-value	p-value	Decision
Total Hardness × TCC	0.262	0.162	$p > 0.05$ (NS)
Total Hardness × <i>E. coli</i>	0.135	0.477	$p > 0.05$ (NS)
Total Hardness × FCC	0.362	0.049	$p < 0.05$ (S)
Iron (Fe) × TCC	−0.239	0.204	$p > 0.05$ (NS)
Iron (Fe) × <i>E. coli</i>	−0.256	0.173	$p > 0.05$ (NS)
Iron (Fe) × FCC	−0.461	0.010	$p < 0.05$ (S)
Lead (Pb) × TCC	0.274	0.142	$p > 0.05$ (NS)
Lead (Pb) × <i>E. coli</i>	0.259	0.167	$p > 0.05$ (NS)
Lead (Pb) × FCC	0.246	0.190	$p > 0.05$ (NS)
TCC × <i>E. coli</i>	0.840	<0.001	$p < 0.05$ (S)
TCC × FCC	0.602	<0.001	$p < 0.05$ (S)
TCC × THBC	0.374	0.041	$p < 0.05$ (S)
TCC × SS Count	0.470	0.009	$p < 0.05$ (S)
<i>E. coli</i> × FCC	0.734	<0.001	$p < 0.05$ (S)
<i>E. coli</i> × THBC	0.499	0.005	$p < 0.05$ (S)
FCC × THBC	0.448	0.013	$p < 0.05$ (S)

TCC = Total Coliform Count; FCC = Fecal Coliform Count; THBC = Total Heterotrophic Bacterial Count; SS = Suspended Solids; S = Significant; NS = Not Significant. Source: Computed from Laboratory Analysis data, 2026.

DISCUSSION

The Household-level information from Isale-Oyo reflect a single contamination chain that runs from source down through the subsequent links in the water-use chain. Fewer than half of the respondents reported that they had treated drinking water before drinking, consistent with Nfor *et al.* (2025) who found comparably limited access to safely managed drinking water and widespread open defecation among Nigerian communities, which they linked to the persistent burden of waterborne disease from microbiologically contaminated drinking water. In this context, where the *E. coli* was present in every sample of the 30 tested sources, source-level contamination can be regarded as universal, underscoring the critical importance of consistent household-level water treatment prior to consumption. The minority who did treat their water most commonly boiled water, followed by filtration, chlorination, and sedimentation, reflecting a hierarchy that aligns with Atalay *et al.* (2024), whose systematic review and meta-analysis found that boiling is among the most commonly reported household water treatment methods in sub-Saharan Africa because of its accessibility and its established ability to inactivate microorganisms. However, the downside of boiling is that it fails to remove chemicals such as chloride, phosphate and iron,

all of which had been found beyond WHO guidelines in this study. The communities can thus be doing an active and passive activity to neutralize the pathogen but at the same time get exposed to the chemistry.

An encouraging development with storage behavior was the presence of covered containers: Nfor *et al.* (2025) similarly found that unsafe household water handling and open defecation were closely linked to bacteriological contamination of stored drinking water. Despite this partial success, sanitation problems were found to be overwhelming, with half of the respondents using open dump for solid waste, pit latrines dominating the methods of excreta disposal and the reported hand washing with soap was limited. The findings are similar to the known correlation between substandard sanitation facilities and contamination of groundwater in Nigeria as reported by Yahaya *et al.* (2021) and Taiwo *et al.* (2023) and consistent with NCDC (2024) surveillance data identifying Oyo State among the states that continue to struggle with active cholera transmission in Nigeria. The Health Belief Model (HBM) (Rosenstock, 1974) provides an interpretation of the reasons behind the persistent gaps (1) residents' low perceived vulnerability to waterborne illness; (2) their lack of connection between symptoms-free water contamination and actual risk; and (3) real or perceived obstacles to obtaining treatment supplies. This type of intervention (targeted risk communication, education that brings the water-disease relationship to life and subsidised access to simple technologies – including ceramic filters or sodium hypochlorite solutions) could change these long-held perceptions.

The physicochemical results indicate a groundwater system that is under geochemical strain. A mean pH of 5.66 represents uniform acidity which agrees with southwestern Nigerian groundwater, with similarly low, acidic pH reported by both Ganiyu *et al.* (2021) and Edeki *et al.* (2023) and attributed to the fact that the region's geology (Precambrian basement) was low in acid-buffering minerals, coupled with CO₂ from the release of organic matter above shallow aquifers. Human activities are likely to accentuate this natural acidity through the addition of anthropogenic sources such as leachate from open dumps, humic acids formed from the decomposition of refuse, and carbonic acid from the infiltration of wastewater. The impact is not only on the aquifer itself, but also on the components of the water system that it flows through, including metal pipes, fittings, and storage units, which can lead to the release of lead from solder or galvanized surfaces into the water. The concentration of chloride was also high with a mean of 432.96 mg/L (2.6 times the permissible limit of 250 mg/L) and a maximum of 639.05 mg/L (2.6 times the permissible limit). Although the chloride concentrations reported in this study did not exceed the threshold for acute toxicity, the presence of these levels is indicative of sewage or urine contamination as found by Balogun *et al.* (2022) who observed that chloride enrichment is a chemical fingerprint of leachate intrusion near settlements in southwestern Nigeria. It is interesting to note that the spatial range of chloride concentrations (106.65–639.05 mg/L) points more towards localized point sources (typically near certain latrines or dump sites) than towards a diffuse geochemical signature of the basin. The story is similar with the exceedances of phosphate, which indicate peri-residential sewage infiltration, agricultural runoff, and the use of detergents, and suggest eutrophication in any surface storage. Interestingly, even with the observed open defecation and dumping, nitrate concentrations stayed relatively low, possibly due to denitrification within the anaerobic portions of the aquifer, dilution from tropical rainy season recharge, and/or the possible prevalence of ammonium as the main nitrogenous contaminant that may have been undetected and quantified in future studies. In themselves the temperature and electrical conductivity values (mean EC: 151.59 µS/cm) were unremarkable but are indicative of a soft, weakly mineralized and low oxygen groundwater environment characteristic of the Basement

Complex terrain which has implications for the survival of the microorganisms and the mobilization of the metals as discussed below.

The most significant discovery of this study is microbiological; *E. coli*, total coliforms and fecal coliforms were found in all 30 samples. If the *E. coli* is detected in all sources, then it is not an isolated incident; this is indicative of not only a continuous presence of fecal loading in the aquifer but also renders the water unsafe for direct consumption according to the WHO (2022) criteria. This is in line with the findings of Yahaya *et al.* (2021) and Nfor *et al.* (2025), which attributed widespread coliform presence in wells and boreholes in Nigeria to poor sanitation and proximity of water sources to sanitation facilities, and in line with the results of Edeki *et al.* (2023), whose acidic and microbiologically compromised ground water was likewise clear in appearance, reinforcing that visual clarity should not be equated with microbiological safety, a message extremely relevant for communities that depend on organoleptic judgement. The high levels of total heterotrophic bacteria (mean 0.62×10^4 cfu/mL) also suggest a well-established bacterial flora that has the potential to contain latent pathogens. The positive correlation between indicator organisms ($r = 0.840$ for TCC $\times$ *E. coli*; $r = 0.734$ for *E. coli* $\times$ FCC, both $p < 0.001$) suggests a common contamination source and transport pathway, which is typical of the sanitary environment in the community. The correlation between total coliforms and suspended solids extends the mechanistic understanding as well: attached bacteria may provide a refuge to particulate matter, protecting the bacteria from the stresses of environment and disinfection, which has direct treatment implications as turbidity removal will precede disinfection to maximize effectiveness. The prevalence of diarrhoea, cholera and typhoid—typical fecal-oral outcomes—matches self-reported illness estimates, though self-report is not a substitute for epidemiological confirmation, but the combination of universal source contamination, limited treatment and high prevalence of illness (37.8% reporting frequent illness) suggests a strong causal story. The findings here support the underpinning warnings by WHO (2023) on microbial contamination of drinking-water as the most significant global threat and the NCDC (2024) surveillance of the burden of water-borne diseases in Nigeria at a local level.

The geochemical problem of iron was a fascinating one to solve. The iron-bearing minerals dissolved reductively under low oxygen conditions as observed are in agreement with Laniyan and Adewumi (2019) who attributed elevated metal concentrations near an exposed dumpsite in southwestern Nigeria to waste-driven groundwater pollution and mineral dissolution processes that can lead to iron enrichment. Interestingly, iron showed a negative correlation with the counts of fecal coliforms, likely because the reducing environment of low oxygen concentrations that is conducive to the mobilization of iron is less favorable than higher oxygen environments for the growth of *E. coli*. Iron may be associated with reductions in the counts of *E. coli*, but not an increase in their safety. In addition to aesthetic issues (discoloration, metal taste) which might encourage residents to use less regulated sources, the pro-oxidant property of iron has raised concerns regarding long-term health effects being investigated (WHO, 2022).

However, lead remained low in all samples compared to WHO guidelines, in contrast to communities with significant industrial lead sources, such as those studied by Yahaya *et al.* (2021), indicating that the sources of lead in the present samples are more geogenic in origin. However, caution is needed: WHO (2022) states that there is no safe level for the neurodevelopmental effects of lead and even conciliatory blood lead concentrations of 2–5 µg/dL have been associated with measurable cognitive deficits (Lanphear *et al.*, 2005). The potential for lead to leach from household plumbing, downstream of a lead-safe source, cannot be ruled out and should be investigated in priority if the pH of the groundwater is in the acidic range and is known to be corrosive.

Taken together, these results indicate a multi-dimensional public health crisis in real time in Isale-Oyo, characterized by universal contamination by faecal micro-organisms, corrosive acidic groundwater, chloride and phosphate contamination reflecting widespread sewage infiltration, aesthetic and geochemical disruption of iron contamination and use of water by household members that exacerbates source-level risk. Hazard identification, as defined in the USEPA (2024) Environmental Health Risk Assessment framework, does confirm concurrent threats of microbiological, physicochemical, and chemical hazards; dose-response and exposure pathways are not fully quantified in this cross-sectional study, but patterns of self-reported illness strongly argue for higher risk characterization for the community. Given the universal contamination of the drinking water supply, its poor treatment, and high prevalence of disease, there is little doubt that the drinking water problem in Isale-Oyo is no abstract number; it's real, quantifiable, and urgent. The overall situation of drinking water in Isale-Oyo indigenous community is a multi-dimensional, concurrent public health crisis.

CONCLUSION

The present study presents the first comprehensive study of the drinking water quality parameters of physicochemical, microbiological and heavy metal levels in the indigenous community of Isale Oyo, as well as the practices of the human hands handling the water, and compares these water quality parameters with the NSDWQ (2015) and WHO (2022) guidelines. The findings reveal a system in crisis at every point. Thirty of the 30 samples collected were microbiologically contaminated, containing *E. coli* in all, and several of the physicochemical parameters were universally above the permissible limit, including pH, chloride, phosphate and iron. The pattern is caused by the combination of the poor local geology, sanitary conditions, open defecation, and indiscriminate disposal of waste, resulting in sewage leaching into shallow aquifers. Lead was in acceptable limits although the acidity of the aquifer does present some cause for concern about the leaching of metals from the distribution infrastructure due to the aging nature of the system. As stated, contamination at the source is not solved at the household level, but rather increased. The laboratory findings were corroborated by the clinical findings as there are behavioural gaps in self-reported morbidity. There are strong positive relationships between each pair of bacteriological indicators, suggesting a common source of fecal pollution, and between hardness, iron, and fecal coliforms, indicating a more complicated geochemical signature for aquifer vulnerability. The evidence when examined as a whole reveal not just a water-quality problem, but a cascading, multi-layer public health challenge that requires prompt and concerted response.

RECOMMENDATIONS

The structural upgrade of the wells and boreholes in the source areas should include concrete apron, raised well-heads (0.5m and above), protective cover, drainage channels, and enforce a 30m sanitary buffer between water sources and sanitation facilities and refuse dumps. Given the fact that a high level of acidity was recorded in the groundwater (pH < 6.5), alkalization using food-grade lime or soda ash is recommended at community scale to bring the groundwater within the WHO range of 6.5–8.5, thus reducing its corrosivity and improving its palatability.

Household Water Treatment and Safe Storage should encourage households to use chlorination, boiling and subsidized ceramic or biosand filters using messaging based on the Household Water Treatment and Safe Storage components of the Health Belief Model. This should be reinforced in the community through health promotion activities and at school through hygiene education focusing on handwashing with soap.

Local governments should increase the efforts to combat open defecation and poor sanitation practices, and roll out pilot CLTS in Isale-Oyo at the governance level. The standard water quality monitoring (quarterly microbiology, annual physicochemical/heavy metal) should be established and reported to the public. Findings should also be used in advocacy to State and Federal water authorities for continued funding of WASH, and future studies should consider seasonal sampling, quantification of ammonium and manganese, and epidemiological case–control studies and evaluation of the extraction of lead from household plumbing.

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The authors declare that they have no competing interests

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Use of Artificial Intelligence

The authors declare that no large language model was used and no data generation, analysis, or interpretation was performed by AI. The authors are fully responsible for the integrity and accuracy of the work.

Authors Contribution

Ayoola Omolola Idowu, Ishola Oluwaseun Adebayo, Olajide Abosede Temitope and Kehinde Samuel Abiodun led the study conception, methodology, data analysis, interpretation, visualization and manuscript drafting. Omotoso Ifeoluwa Omotola, Abiola Busola Tolulope, Ogunmola Patricia Olusayo, Ola Victor Olabode, and Oladoye Rukayat Olawumi contributed to the literature review, interpretation of findings and manuscript revision. Olajide Abosede Temitope, Oladoye Rukayat Olawumi, Awolaye Sunday Joseph, and Kehinde Samuel Abiodun contributed to the critical review of the manuscript and revision of the final draft. All authors reviewed and approved the final manuscript.

Availability of Data and Materials

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

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